EV Adoption Around the World

Modelling with the S-Curve

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How can we calculate who's next?

EV-Mobility-Modelling

A Comparative Study of S-Curve Modelling of EV Adoption and the Role of National Policies

Climate Change Calculated, TU Ilmenau

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Motivation: Why is this debate happening now?

Why is the automobile industry trying to water down the combustion-engine phase-out in 2035,
even though it is still about 10 years in the future —
and countries like Norway have almost already reached this goal?

Motivation: Why is this debate happening now?

To answer this, we need to address three questions:

- How can we predict and estimate the coming years?
- How do we judge the current situation?
- Which signals matter — and which are just noise?

We need a method that is simple, transparent, and based on data — not wishful thinking.

The solution: Modeling technology adoption

A powerful way to understand technological change is to look at past transitions.

From horses to cars, from analog to digital photography, from simple mobile phones to smartphones — the adoption follows a characteristic S-shaped curve.

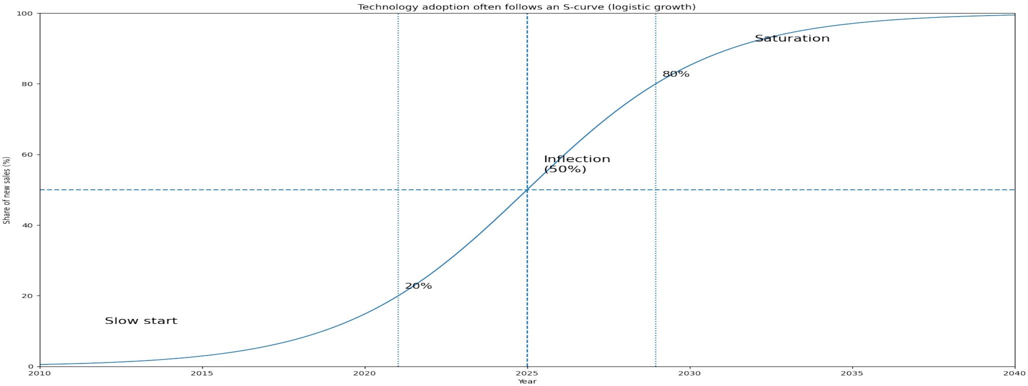
The S-curve is a surprisingly powerful yet simple model, with only a few parameters:

- the growth rate
- the final saturation level
- and the tipping point in time

Policies can influence all of these parameters — but they cannot change the basic S-shape itself.

Most technologies grow like an S

- Slow start $\rightarrow$ inflection (tipping point) $\rightarrow$ saturation.
- Shares are better than raw counts for comparing countries.



The logistic model

$$s(t) = \frac{L}{1 + e^{-k(t-t_0)}}$$

- L: saturation (max share).
- k: growth speed.
- t_0: inflection year (50%).

We just need to determine these parameters such that the curve fits the observed data points.

Fitting the logistic model

The S-curve can

- be early or late, with t_0
- have slow or fast growth, with k
- have less than 100% saturation, with L

Open data (reproducible)

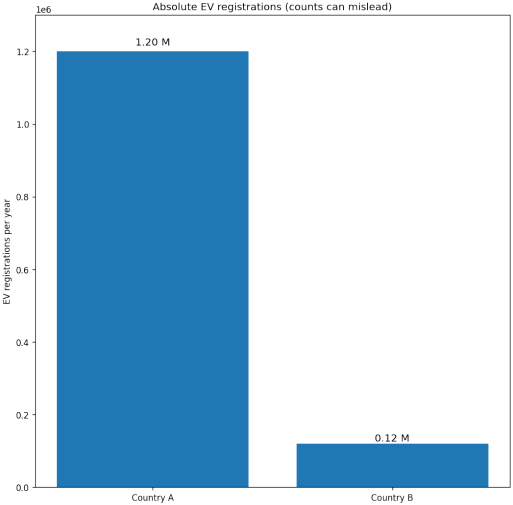
- Global: IEA Global EV Data Explorer
- Hannah Ritchie (2024) - "Tracking global data on electric vehicles" Published online at OurWorldinData.org. Retrieved from: `https://archive.ourworldindata.org/20251208-170037/electric-car-sales.html` [Online Resource] (archived on December 8, 2025).
- More sources in the description and in: `https://github.com/TUIlmenauAMS/ClimateChangeCalculated`

Open data (reproducible)

- All plots are generated from CSVs and code linked in the description.
- Code was written with the help of our **"Climate Change Assistant"**.
- This assisant can also be used at any time for background questions `https://chatgpt.com/g/g-6829ff3e69cc81918c917c75c6925b35-climate-change-assistant`

What counts as an EV?

- BEV: Battery Electric Vehicle
- PHEV: Plug-in Hybrid
- Main model: BEV+PHEV share of new registrations

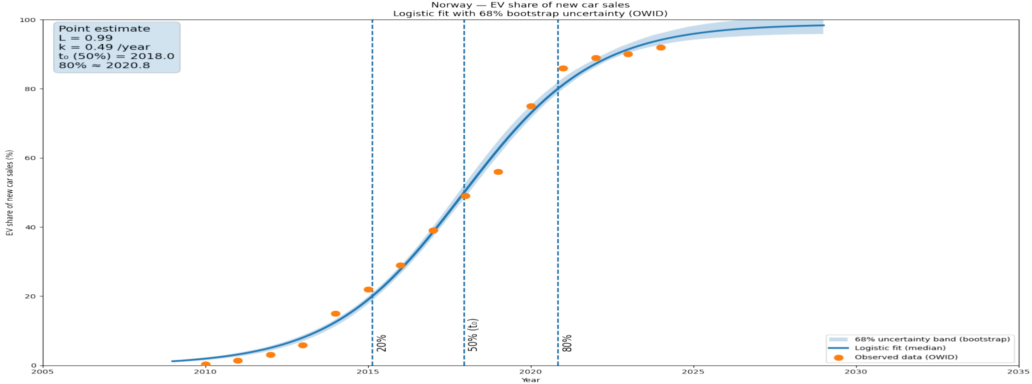


Fitting S-curves (what we do in Python)

- Compute BEV+PHEV share $= \frac{\text{BEV+PHEV regs}}{\text{total regs}}$.
- Fit logistic curve with constraints: $0.5 \leq L \leq 1.05, \ k > 0$.
- Solve milestone years for 20%, 50%, 80%.
- **Bootstrap resampling** to get 68% uncertainty bands ($\pm$ standard deviation).
- The shaded plot area shows the **uncertainty bands** of the fitted S-curve.

Example: Norway (late stage)

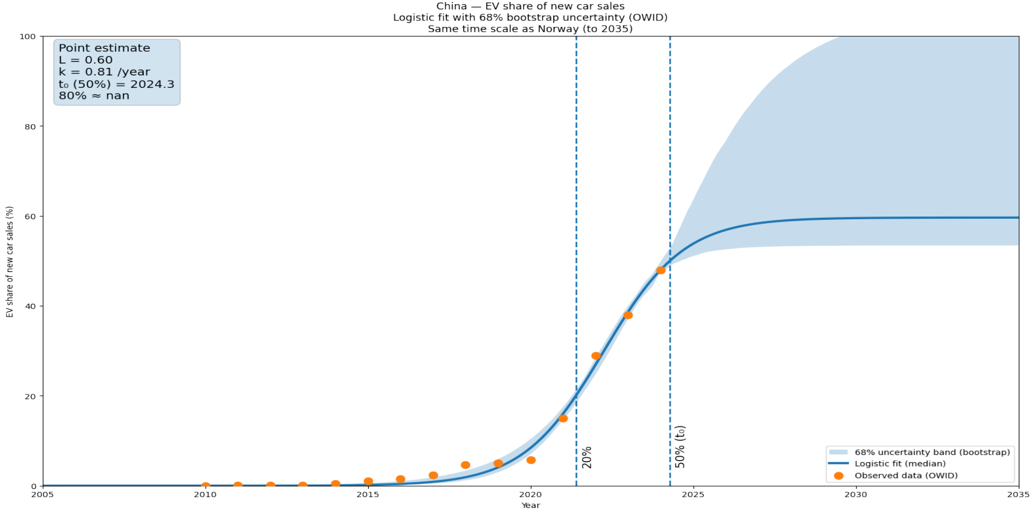
Near saturation: the curve must flatten. The shaded area is the uncertainty band.



Show milestones: t_{20}, t_{50}, t_{80} (median + 68% CI).

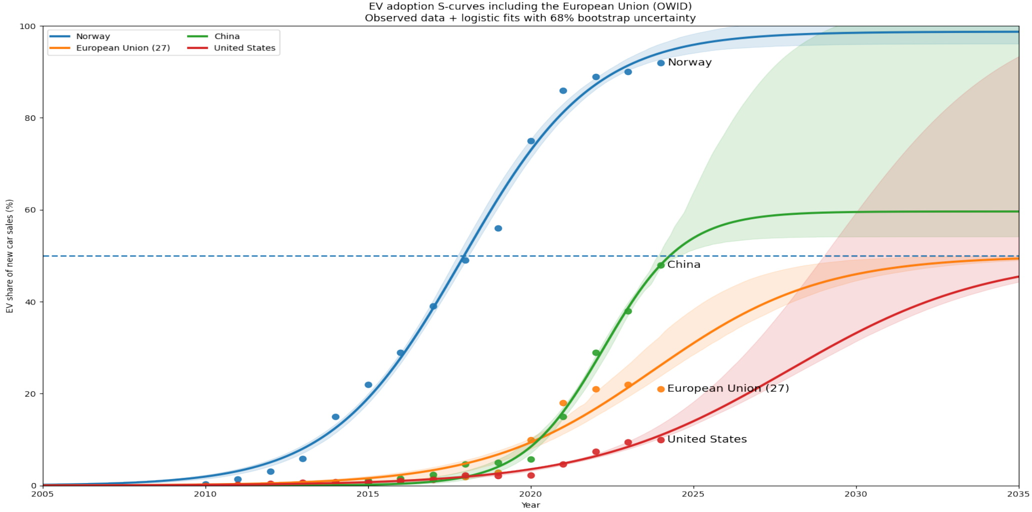
Example: China (mid stage)

Huge market, strong mid-curve acceleration. More uncertainty towards the future.



Multi-country comparison

Different positions on the same S-pattern. The EU converges to low saturation.



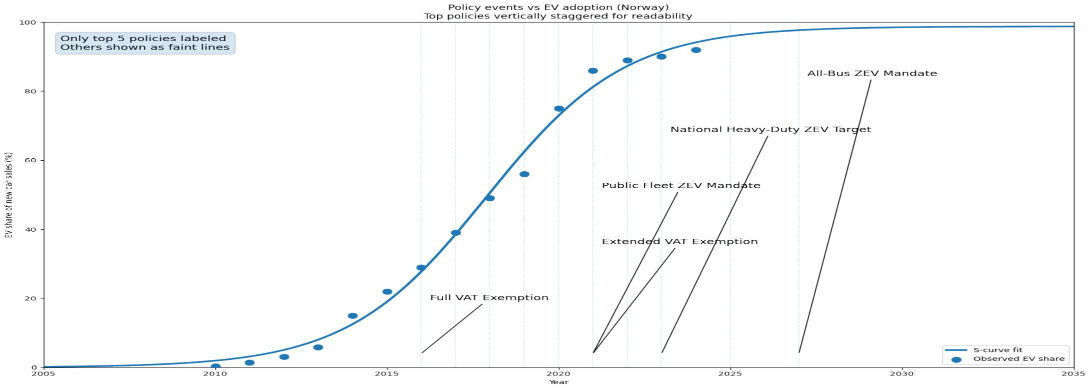
The 2035 Goal of 100% EV

Observe:

- Norway as almost there.
- China has a good chance of reaching this goal.
- Even the US has a realistic chance of reaching this goal, based on current data,
- but Europe appears to be approaching 50% saturation, which means it will be difficult to achieve this goal, and new **policy measures** will be necessary.
- Hence the attempt by the European automotive industry to water down the target.

How policy interacts with the EV S-curve: Norway

Policy does not create adoption — it shifts and accelerates it.

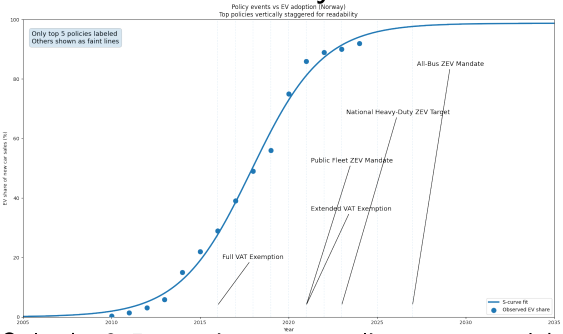


Only the most important policy interventions are labeled; others are shown as faint lines.

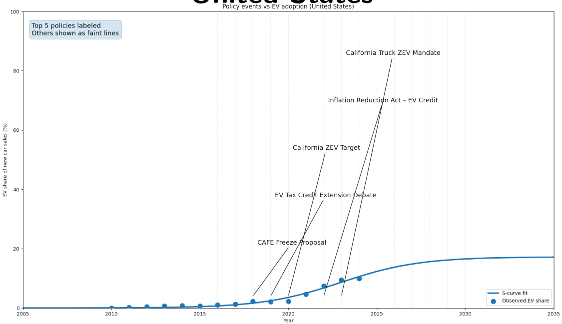
Policy and the EV S-curve: Norway vs United States

Norway applied strong incentives much earlier than the US.

Norway



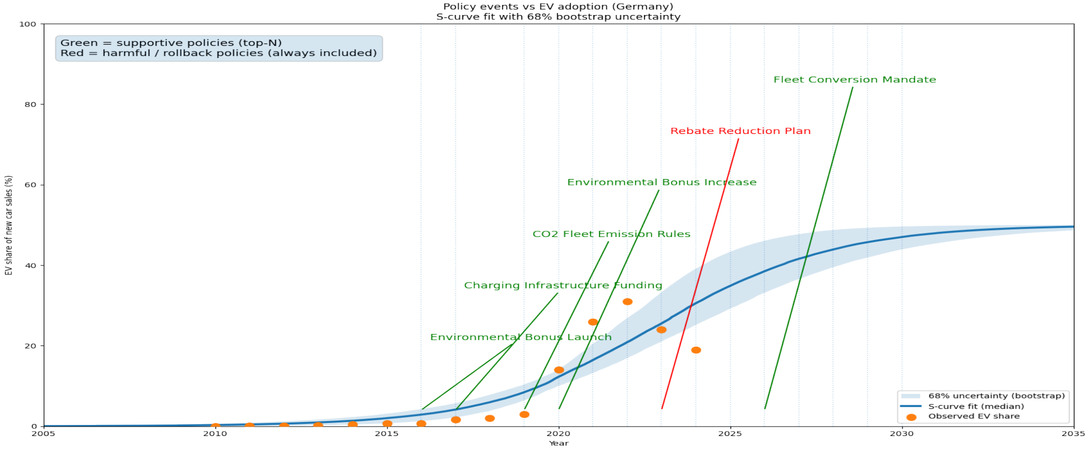
United States



Only the 3–5 most important policy events are labeled; others are shown as faint vertical lines.

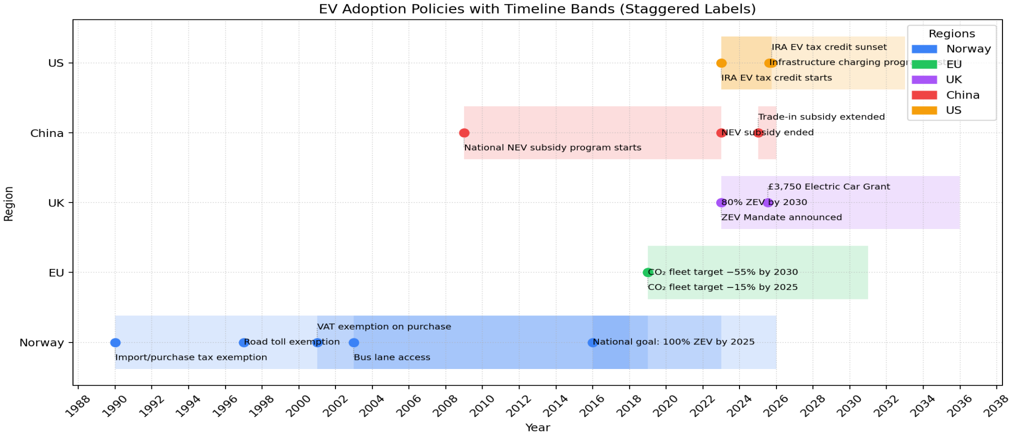
Germany: Policy events and EV adoption uncertainty

Germany is still early on the S-curve, hence policy consistency is important this stage.



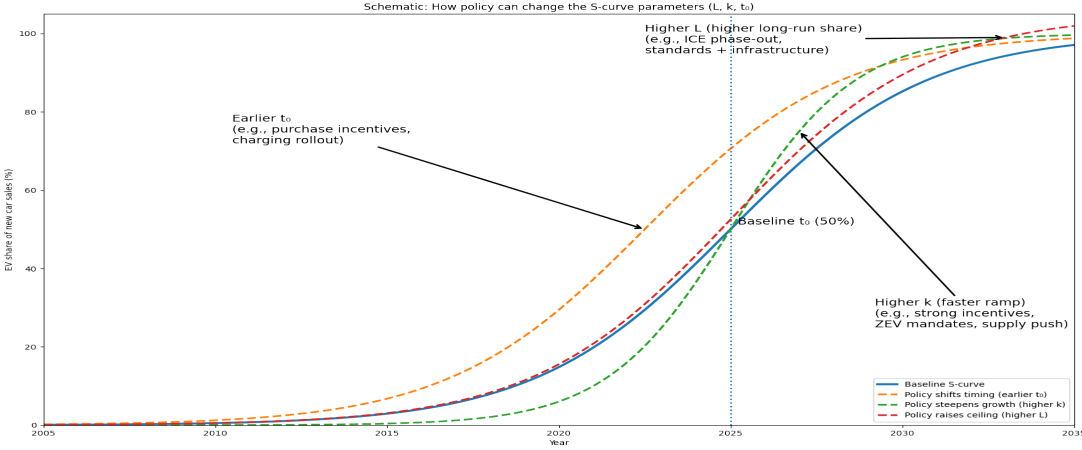
Internation Policy Timelines

- Policy shocks can change k quickly.
- Supply constraints and charging rollout matter.



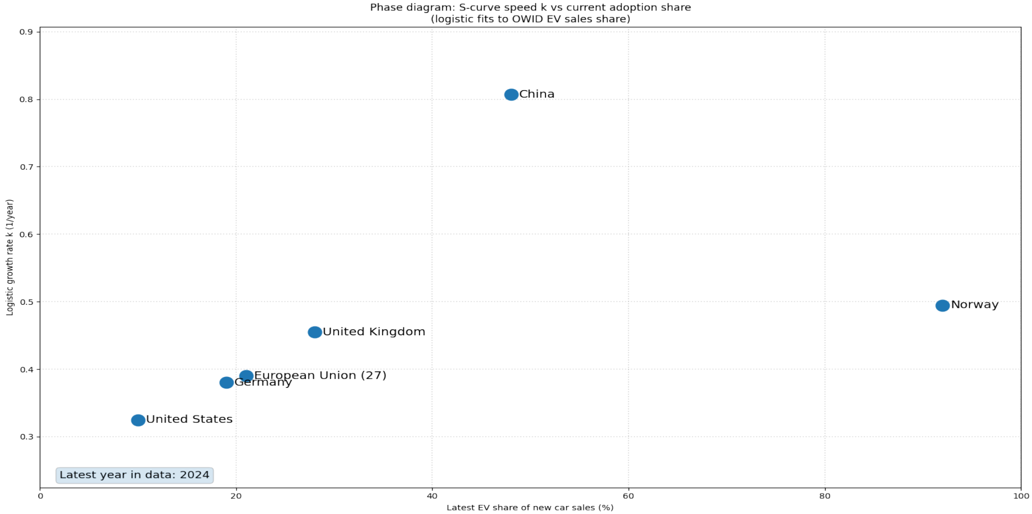
Schematic: How policy changes the S-curve (L, k, t_0)

Policy can shift timing (t_0), steepen growth (k), or raise the ceiling (L).



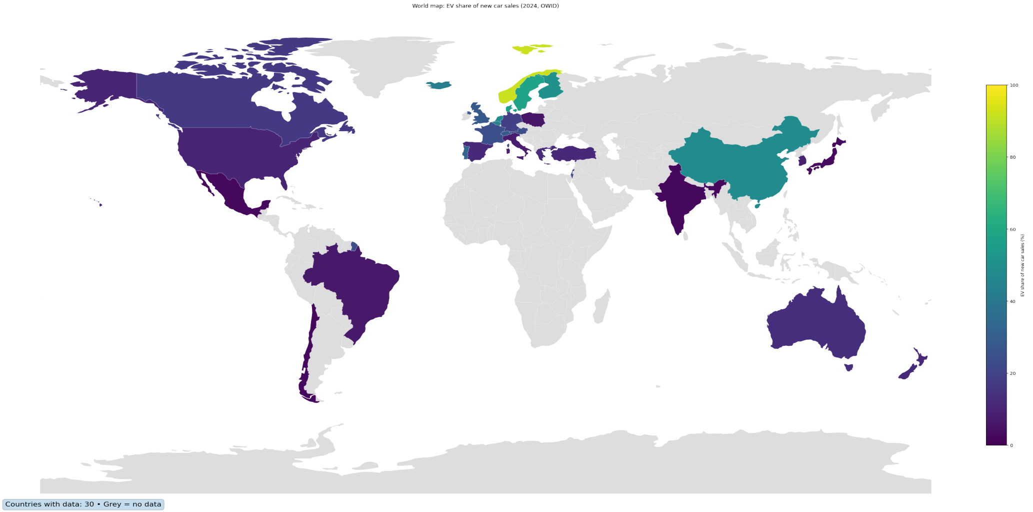
Phase diagram: level vs speed

Who is early-but-fast vs late-and-slow?



World map: current BEV share

Where is adoption already high?



Takeaways

- Many countries are entering the steep part of the S-curve.
- Falling EV costs and better charging push adoption faster.
- Uncertainty is explicit: we show confidence bands.

Data beats opinions.

Conclusions

- The S-curve allows us to estimate the current development dynamics and extrapolate them into the future.
- Political measures influence the start, the speed of the dynamics, and the saturation at the end.
- The EU's current S-curve indicates insufficient saturation, suggesting that supportive policy measures are needed (purchase incentives, legal concessions, combustion engine phase-out, infrastructure, etc.).

Call to action

Try your own country: code + data in the description.

- Comment: who do you think is next?
- Subscribe for more reproducible calculations.